

ASR from Scratch

AppliedNLP mini-project

학생 최재원 | jaeewon24@hufs.ac.kr

Division of Language & AI, Hufs, Seoul

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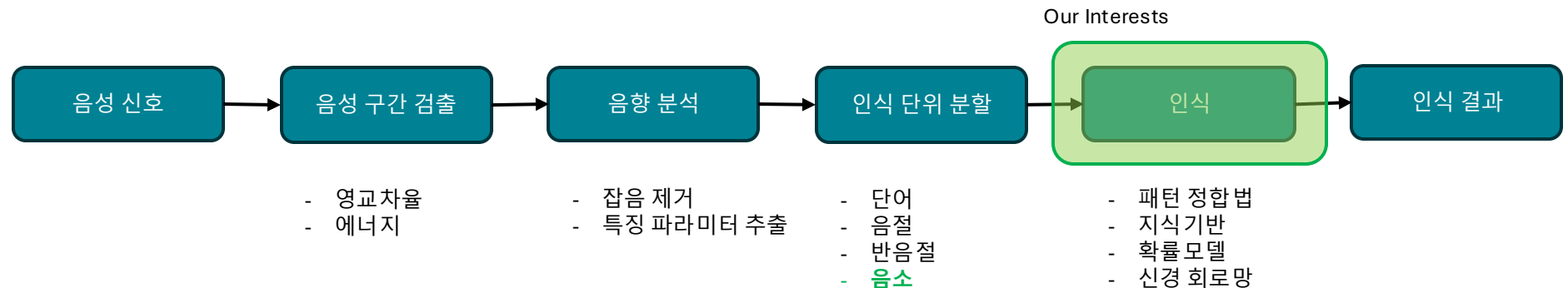
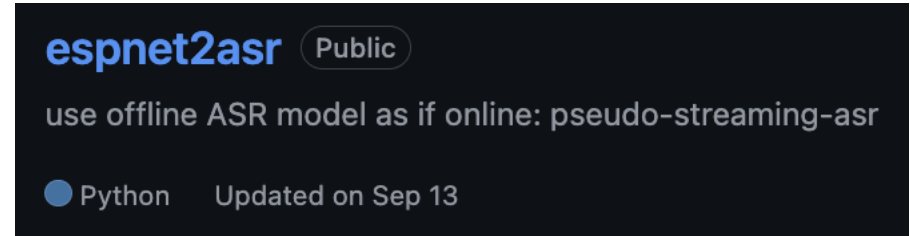
Future Works

follow-up studies for 'espnet2asr'

phoneme sequence를 입력으로 가질 때
grapheme sequence를 출력으로 하는 **Phoneme to Grapheme** 모델

[Phoneme] K AE1 T -> [Grapheme] cat

Machine Translation 문제로 바라볼 수 있음



1. base-line보다 더 나은 성능을 보여야
2. 동음이의어를 잘 처리할 수 있어야

Baseline 1: lexicon; exact-matching

구현 방식

cmu dict을 순회하며 cmu_dict⁻¹(즉, inversed cmu dict)를 구축
추론 시점에 inversed cmu dict에서 grapheme을 가져오며, OOV 단어들은 <UNK>로 대체하도록

한계

동음이의어의 경우, CMU dict에서 먼저 나오는 단어로 처리되는 한계를 가지고 있음 (아래의 빨간색은 중복되는 발음이 존재함을 의미)

예시

Gold THE STOP AT QUEENSTOWN THE TEDIOUS PASSAGE UP THE MERSEY WERE THINGS THAT HE NOTED DIMLY THROUGH HIS GROWING IMPATIENCE

WB (o) THE STOP AT <UNK> THE TEDIOUS PASSAGE UP THE MERCY WERE THING'S THAT HE NOTED DIMLY THREW HIS GROWING IMPATIENCE

WB (x) <UNK>

** 보통은 각 phoneme에 대해 여러 후보들을 두고 LM과 Viterbi를 이용하지만, 미니 프로젝트를 감안해주시면 감사드리겠습니다. **

LibriSpeech; AN ASR CORPUS BASED ON PUBLIC DOMAIN AUDIO BOOKS

- Large-scale (1000 hours) corpus of read English speech
- License: CC BY 4.0

train.clean.100 28,539 rows

train.clean.360 104,014 rows

train.other.500 148,688 rows

음성 분야에서 널리 사용되는 LibriSpeech 데이터셋을 이용

우리의 목적인 Phoneme to Grapheme에 맞도록 후처리가 필요했음



IEEE Xplore

<https://ieeexplore.ieee.org/document>

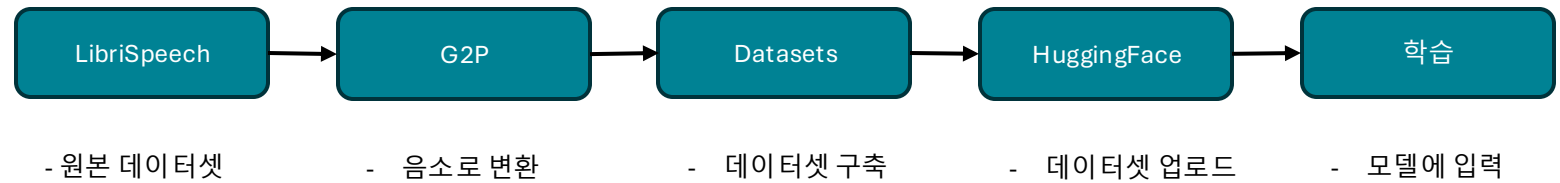
Librispeech: An ASR corpus based on public domain audio ...

V Panayotov 저술 · 2015 · 9045회 인용 — This paper introduces a new corpus of **read English speech**, suitable for training and evaluating speech recognition systems.

Datasets; jaeewon/librispeech_phonemes

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train set의 transcription을 G2P를 이용해 CMU-dict 형식의 phonemes로 변환하여, 이를 HuggingFace에 업로드



Commit **cb2bcb9**
jaeewon committed on Oct 28

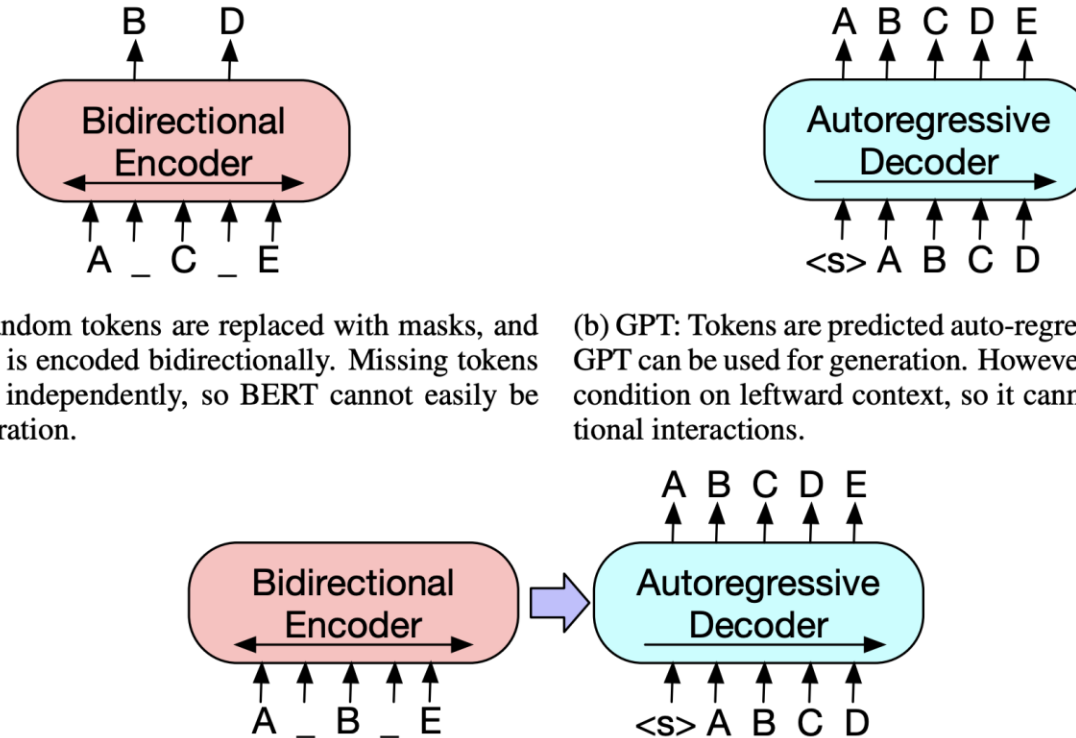
added: uploaded librispeech_phonemes to hf datasets

master

Hugging Face 🔍 Search models, datasets, users...

Datasets: jaeewon/**librispeech_phonemes** ❤️ like 0

id	text	phonemes
string · lengths	string · lengths	list · lengths
14 16	3 516	3 454
1673-143397-0000	ARDENT IN THE PROSECUTION OF HERESY CYRIL AUSPICIOUSLY OPENED HIS REIGN BY OPPRESSING THE NOVATIANS THE MOST...	["AA1", "R", "D", "AH0", "N", "T", " ", " ", "IH0", "N", " ", " ", "DH", "AH0", " ", " ", "P", "R", "AA2", "S", ...]
1673-143397-0001	WITHOUT ANY LEGAL SENTENCE WITHOUT ANY ROYAL MANDATE THE PATRIARCH AT THE DAWN OF DAY LED A SEDITION...	["W", "IH0", "TH", "AW1", "T", " ", " ", "EH1", "N", "IY0", " ", " ", "L", "IY1", "G", "AH0", "L", " ", " ", "S" ...]
1673-143397-0002	SUCH CRIMES WOULD HAVE DESERVED THE ANIMADVERSION OF THE MAGISTRATE BUT IN THIS PROMISCUOUS OUTRAGE THE...	["S", "AH1", "CH", " ", " ", "K", "R", "AY1", "M", "Z", " ", " ", "W", "UH1", "D", " ", " ", "HH", "AE1", "V", ...]
1673-143397-0003	THE ZEAL OF CYRIL EXPOSED HIM TO THE PENALTIES OF THE JULIAN LAW BUT IN A FEEBLE GOVERNMENT AND A...	["DH", "AH0", " ", " ", "Z", "IY1", "L", " ", " ", "AH1", "V", " ", " ", "S", "IH1", "R", "AH0", "L", " ", " ", "IH0" ...]
1673-143397-0004	ORESTES COMPLAINED BUT HIS JUST COMPLAINTS WERE TOO QUICKLY FORGOTTEN BY THE MINISTERS OF THEODOSIUS AND...	["A00", "R", "EH1", "S", "T", "IY0", "Z", " ", " ", "K", "AH0", "M", "P", "R", "EH1", "N", "D", " ", ...]
1673-143397-0005	A RUMOR WAS SPREAD AMONG THE CHRISTIANS THAT THE DAUGHTER OF THEON WAS THE ONLY OBSTACLE TO THE...	["AH0", " ", " ", "R", "UW1", "M", "ER0", " ", " ", "W", "AA1", "Z", " ", " ", "S", "P", "R", "EH1", "D", " ", ...]



(a) BERT: Random tokens are replaced with masks, and the document is encoded bidirectionally. Missing tokens are predicted independently, so BERT cannot easily be used for generation.

(b) GPT: Tokens are predicted auto-regressively, meaning GPT can be used for generation. However words can only condition on leftward context, so it cannot learn bidirectional interactions.

(c) BART: Inputs to the encoder need not be aligned with decoder outputs, allowing arbitrary noise transformations. Here, a document has been corrupted by replacing spans of text with mask symbols. The corrupted document (left) is encoded with a bidirectional model, and then the likelihood of the original document (right) is calculated with an autoregressive decoder. For fine-tuning, an uncorrupted document is input to both the encoder and decoder, and we use representations from the final hidden state of the decoder.

Figure 1: A schematic comparison of BART with BERT (Devlin et al., 2019) and GPT (Radford et al., 2018).

특징

Pretraining stage (1) text is corrupted with an arbitrary noising function

Pretraining stage (2) a sequence-to-sequence model is learned to reconstruct the original text.

A key advantage of this setup is the noising flexibility; arbitrary transformations can be applied to the original text, including changing its length.

학습 데이터

160Gb of news, books, stories, and web text.

모델 선택 이유

우리의 목표와 유사하게 학습됨
우리의 목표를 수행할 수 있을 정도의 견고한 모델 구조
우리의 목표에 적절한 모델 구조
우리의 목표를 쉽게 구현할 수 있는 모델 구조

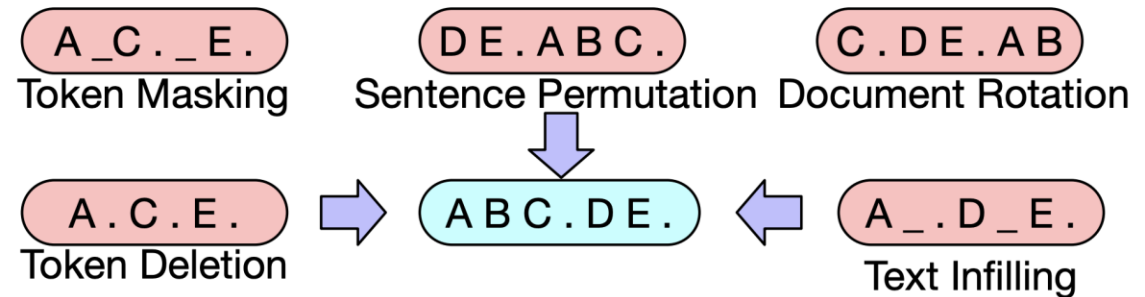


Figure 2: Transformations for noising the input that we experiment with. These transformations can be composed.

Why **BartForConditionalGeneration** ?

우리가 풀 문제는 전형적인 Seq2Seq 문제

decoder-only인 BartForCausalLM은 적합하지 않음

Encoder + Decoder 모두 사용하며 Decoder에서 Cross-Attention을 사용해야

```
modeling_bart.py ×
home > jpong > miniconda3 > envs > applied_nlp > lib > python3.11 > site-packages > transformers > models > bart > modeling_bart.py > BartForConditionalGeneration
1558 > @add_start_docstrings( ...
1561 > class BartForConditionalGeneration(BartPreTrainedModel, GenerationMixin): ...
1698
1699
1700 > @add_start_docstrings( ...
1707 > class BartForSequenceClassification(BartPreTrainedModel): ...
1828
1829
1830 > @add_start_docstrings( ...
1837 > class BartForQuestionAnswering(BartPreTrainedModel): ...
1953
1954
1955 > class BartDecoderWrapper(BartPreTrainedModel): ...
1967
1968
1969 > @add_start_docstrings( ...
1975 > class BartForCausalLM(BartPreTrainedModel, GenerationMixin): ...
2161
```

learning_rate: 1e-4

per_device_train_batch_size: 32

per_device_eval_batch_size: 48

num_train_epochs: 30

warmup_ratio: 0.1

weight_decay: 0.01

bf16: True

gradient_accumulation_steps: 1

label_smoothing_factor: 0.1

처음에 계획했던 "phonemes에 노이즈를 섞는 전처리"는 **하지 않음**

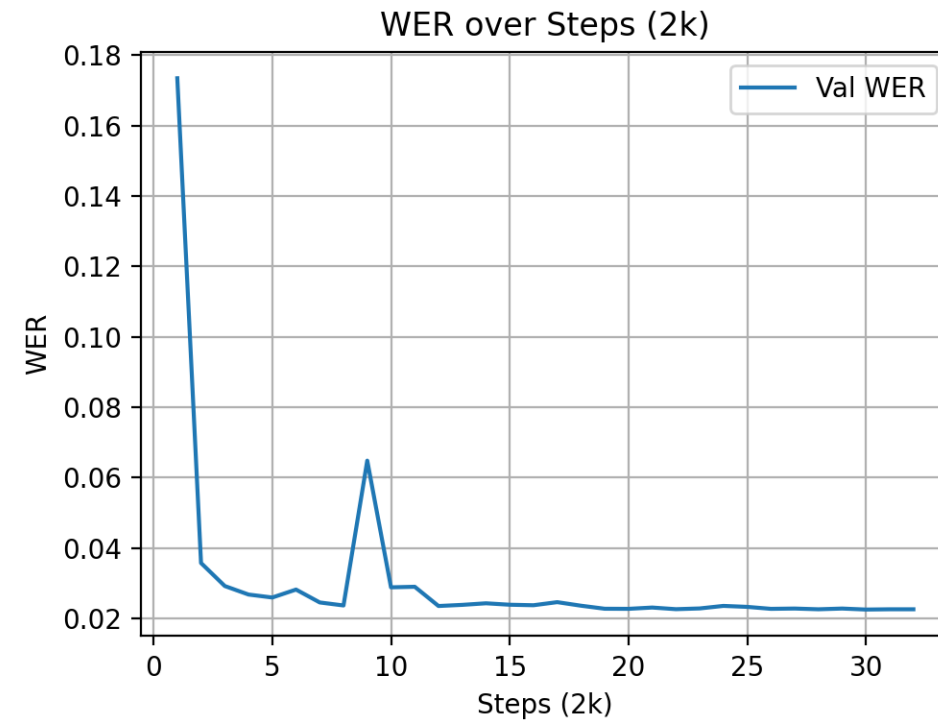
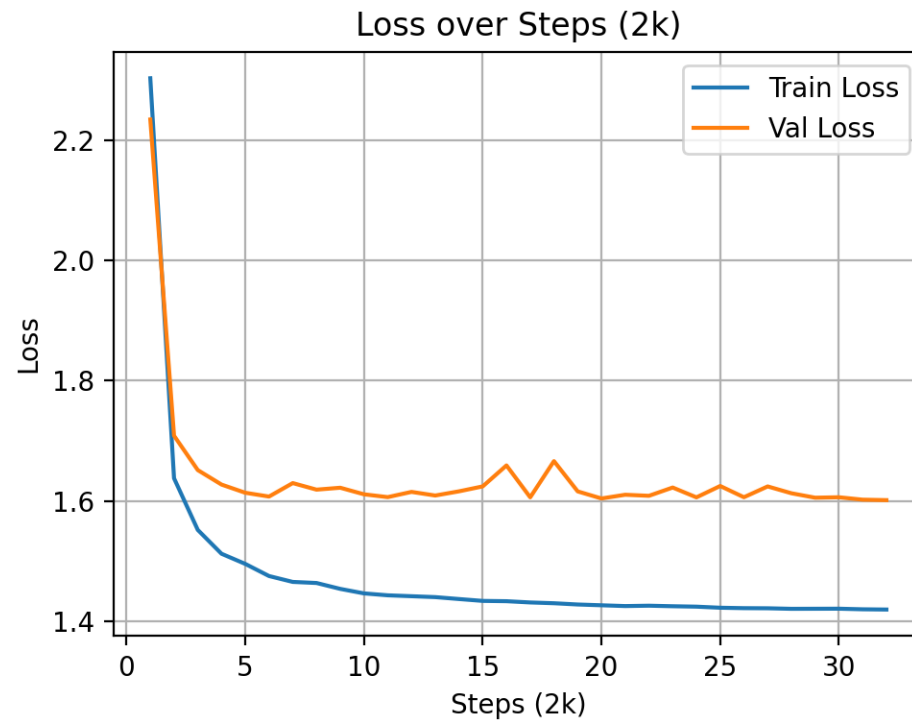
대신, 단어간 경계를 모델에게 명시적으로 알려줬을 때와 그렇지 않을 때를 비교함

단어 경계 o: RTX5090 4대로 7시간 4분 동안 학습

단어 경계 x: RTX5090 4대로 6시간 37분 동안 학습

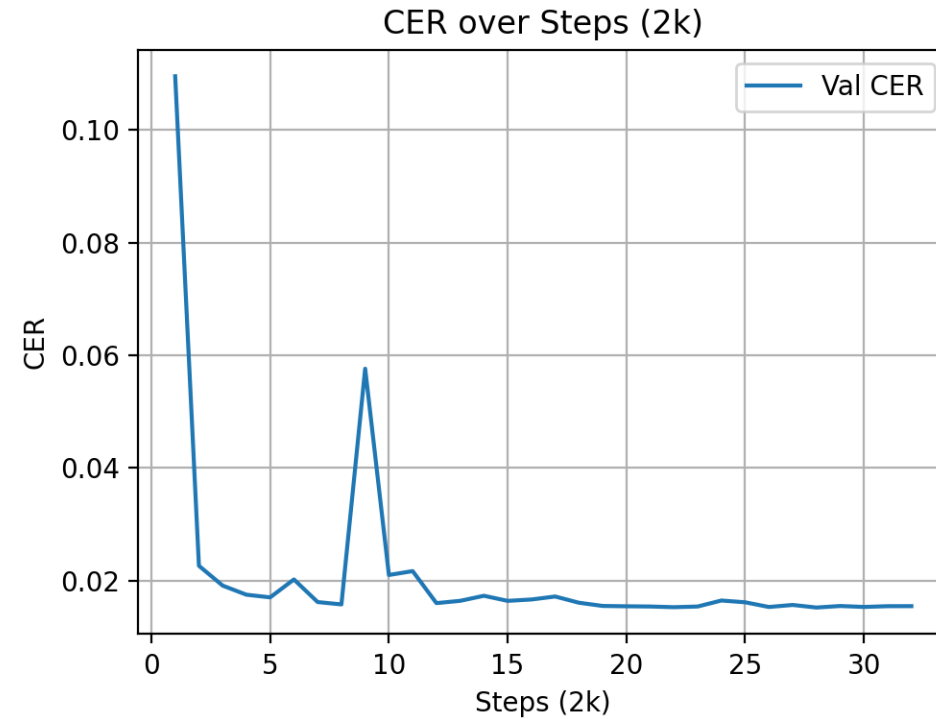
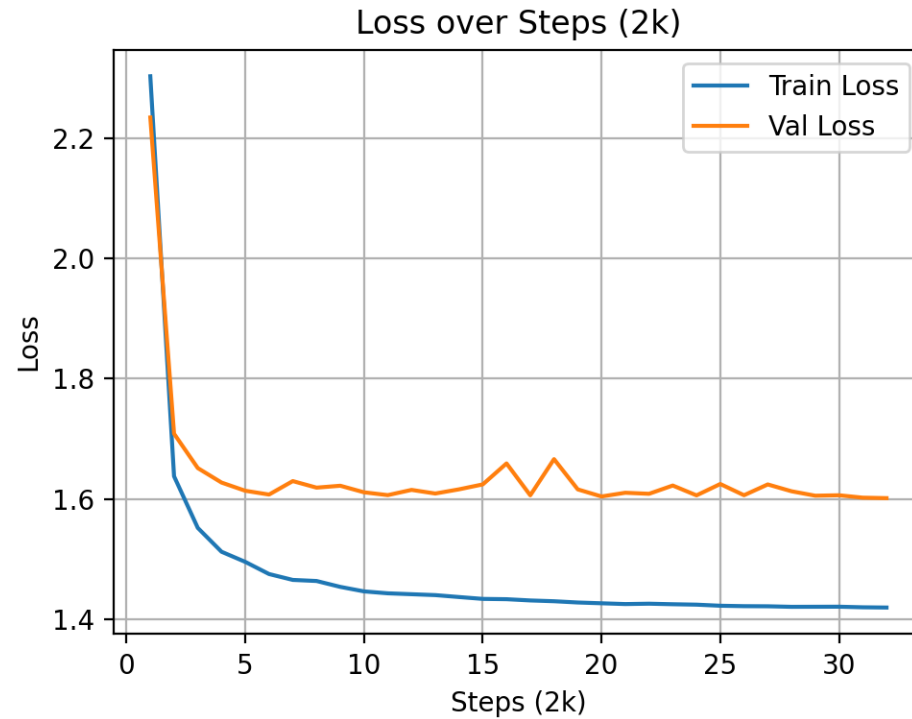
Evaluate & Analysis in Validation Set (WER)

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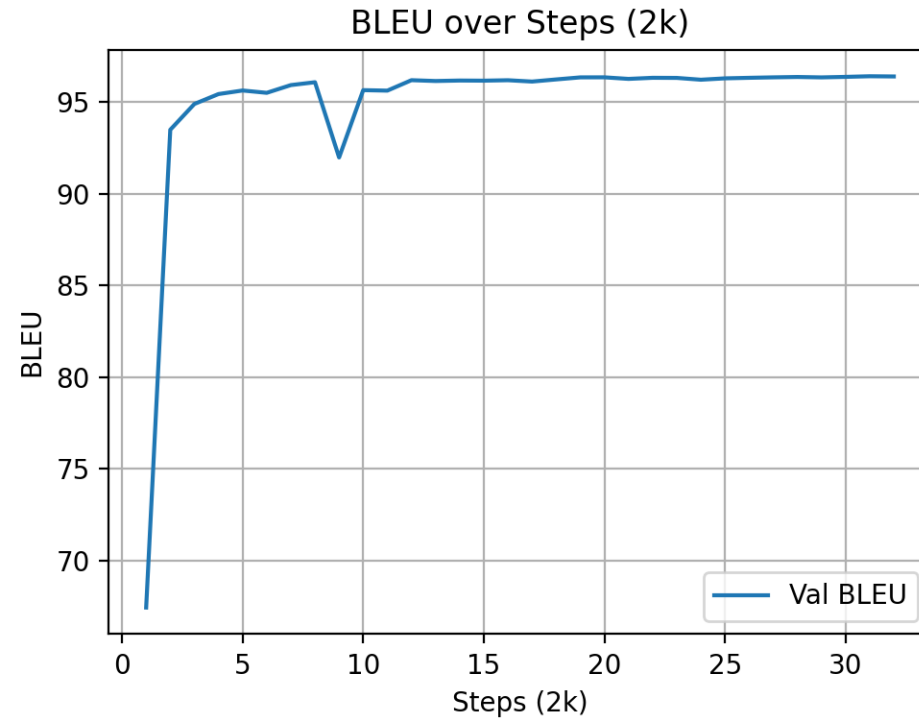
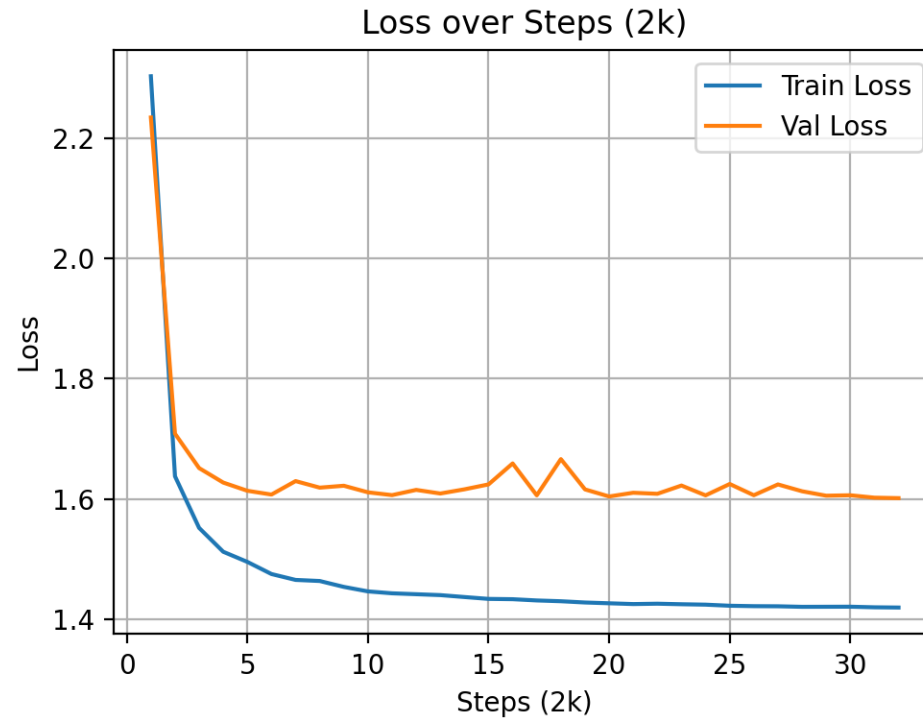
Evaluate & Analysis in Validation Set (CER)

13



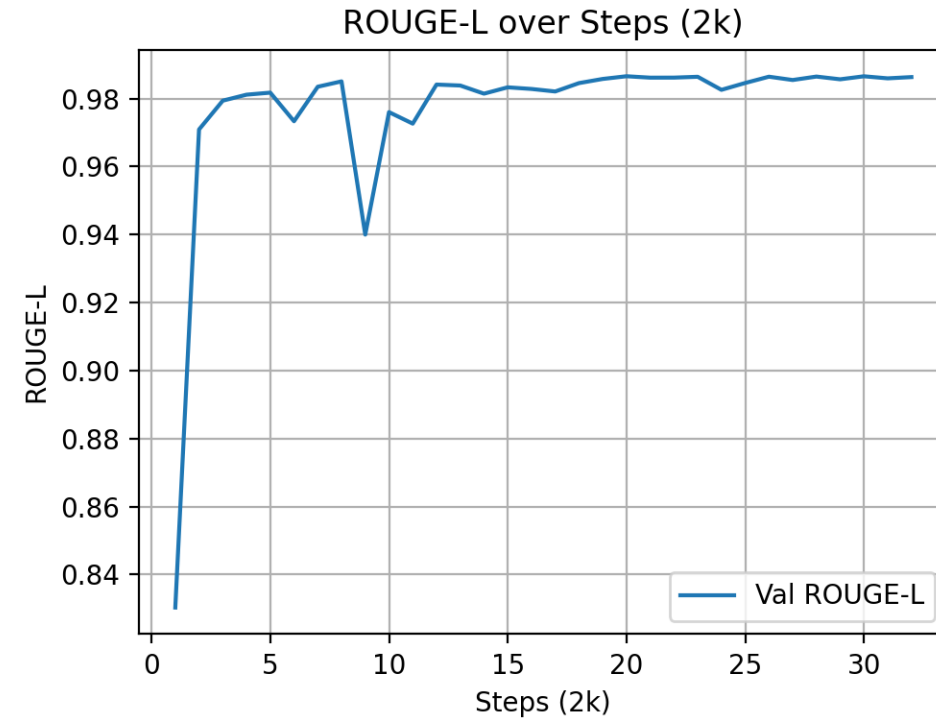
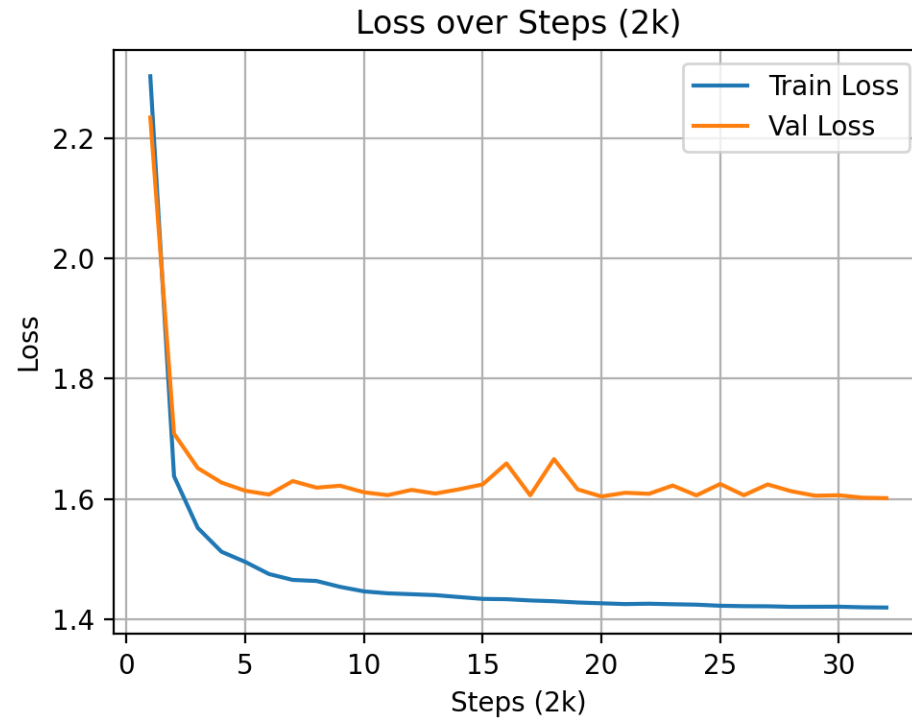
Evaluate & Analysis in Validation Set (BLEU)

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Evaluate & Analysis in Validation Set (ROUGE-L)

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===== results for dev.clean =====

dev.clean BLEU : 47.3316

dev.clean WER : 0.2562

dev.clean CER : 0.1028

dev.clean ROUGE1 : 0.7456

dev.clean ROUGE2 : 0.5570

dev.clean ROUGEL : 0.7452

===== results for test.clean =====

test.clean BLEU : 47.4273

test.clean WER : 0.2579

test.clean CER : 0.1029

test.clean ROUGE1 : 0.7448

test.clean ROUGE2 : 0.5614

test.clean ROUGEL : 0.7442

===== results for dev.other =====

dev.other BLEU : 44.1307

dev.other WER : 0.2788

dev.other CER : 0.1161

dev.other ROUGE1 : 0.7171

dev.other ROUGE2 : 0.5222

dev.other ROUGEL : 0.7165

===== results for test.other =====

test.other BLEU : 44.1736

test.other WER : 0.2776

test.other CER : 0.1154

test.other ROUGE1 : 0.7211

test.other ROUGE2 : 0.5286

test.other ROUGEL : 0.7205

macro avg.
BLEU: 45.7658
WER: 0.2674
CER: 0.1093

===== results for dev.clean =====

dev.clean BLEU : 96.3666

dev.clean WER : 0.0244

dev.clean CER : 0.0178

dev.clean ROUGE1 : 0.9877

dev.clean ROUGE2 : 0.9775

dev.clean ROUGEL : 0.9877

===== results for test.clean =====

test.clean BLEU : 96.3993

test.clean WER : 0.0245

test.clean CER : 0.0182

test.clean ROUGE1 : 0.9880

test.clean ROUGE2 : 0.9788

test.clean ROUGEL : 0.9880

===== results for dev.other =====

dev.other BLEU : 96.4626

dev.other WER : 0.0204

dev.other CER : 0.0125

dev.other ROUGE1 : 0.9854

dev.other ROUGE2 : 0.9738

dev.other ROUGEL : 0.9854

===== results for test.other =====

test.other BLEU : 95.8044

test.other WER : 0.0256

test.other CER : 0.0167

test.other ROUGE1 : 0.9840

test.other ROUGE2 : 0.9714

test.other ROUGEL : 0.9840

macro avg.
BLEU: 96.2582
WER: 0.0237
CER: 0.0163

===== results for dev.clean =====

dev.clean BLEU : 97.0897

dev.clean WER : 0.0164

dev.clean CER : 0.0086

dev.clean ROUGE1 : 0.9883

dev.clean ROUGE2 : 0.9779

dev.clean ROUGEL : 0.9882

===== results for test.clean =====

test.clean BLEU : 97.0976

test.clean WER : 0.0159

test.clean CER : 0.0076

test.clean ROUGE1 : 0.9878

test.clean ROUGE2 : 0.9772

test.clean ROUGEL : 0.9878

===== results for dev.other =====

dev.other BLEU : 96.8522

dev.other WER : 0.0163

dev.other CER : 0.0067

dev.other ROUGE1 : 0.9853

dev.other ROUGE2 : 0.9738

dev.other ROUGEL : 0.9853

===== results for test.other =====

test.other BLEU : 96.2154

test.other WER : 0.0212

test.other CER : 0.0104

test.other ROUGE1 : 0.9827

test.other ROUGE2 : 0.9692

test.other ROUGEL : 0.9827

macro avg.
BLEU: 96.8137
WER: 0.0175
CER: 0.0083

I can **see** the ships sailing on the deep blue **sea**.

My **son** loves to play outside in the **sun**.

Please **write** your name on the **right** side of the paper.

Can you **hear** the music clearly from **here**?

The **two** of us went **to** the park, and she wanted **to** go **too**.

The brave **knight** rode his horse into the dark **night**.

She bought a new **pair** of shoes and ate a juicy **pear**.

He felt very **weak** for a whole **week** after the flu.

Gold : i can **see** the ships sailing on the deep blue **sea**

Pred : i can **see** the ships sailing on the deep blue **sea**

Gold : my **son** loves to play outside in the **sun**

Pred : my **son** loves to play outside in the **sun**

Gold : please **write** your name on the **right** side of the paper

Pred : please **write** your name on the **right** side of the paper

Gold : can you **hear** the music clearly from **here**

Pred : can you **hear** the music clearly from **here**

Gold : the **two** of us went **to** the park and she wanted **to** go **too**

Pred : the **two** of us went **to** the park and she wanted **to** go **too**

Gold : the brave **knight** rode his horse into the dark **night**

Pred : the brave **knight** rode his horse into the dark **night**

Gold : she bought a new **pair** of shoes and ate a juicy **pear**

Pred : she bought a new **pair** of shoes and ate a juicy **pair**

Gold : he felt very **weak** for a whole **week** after the **flu**

Pred : he felt very **weak** for a whole **week** after the **flue**

test set에 대해 WER, CER, BLEU, Rouge-L의 모든 지표에서 이미 충분히 수렴함을 확인

early-stopping 등으로 불필요한 학습을 막아도 좋았겠음

단어간 경계를 모델에게 명시적으로 알려주지 않더라도 충분히 큰 모델(facebook/bart-large)는 알아서 잘 학습함

phoneme인 "T0"를 "<PH_T0>"의 형태로 vocab에 추가하여 학습할 때는 수렴이 안 되었음
그러나, "T0"를 그대로 vocab에 추가하여 학습했더니 수렴함

이것이 원인이 되어 단어간 경계를 <WB>로 모델에게 명시적으로 제공했을 때 제대로 학습이 되지 않았던 것 같음

```
The new embeddings will be initialized from a multivariate normal distribution that has old embeddings' mean and covariance.  
As described in this article: https://nlp.stanford.edu/~johnhew/vocab-expansion.html. To disable this, use `mean_resizing=False`
```

1. 더 강력한 Baseline과의 비교
ex. Viterbi + n-gram
2. 더 작은 모델에서의 가능성 확인
아주 짧게 학습했을 때도 나쁘지 않은 성능을 보였기 때문에, 덜 복잡한 모델에서도 잘 작동할 것임
3. Related Works 조사
가장 먼저 수행했어야 하나, 미니 프로젝트라 건너 뛴
4. ESPnet toolkit을 이용해 학습시킨 모델을 포함하는 ASR system 완성해보기
본 과목을 수강하게 된 이유였기도 함
5. 처음에 계획했던 대로 phonemes에 노이즈를 섞어 학습해 비교

발표에 사용된 모든 자료는 아래 레포지터리에서 확인하실 수 있습니다.

https://github.com/jaeewon/applied_nlp



한국외국어대학교
HANKUK UNIVERSITY OF FOREIGN STUDIES

최재원

꼬꼬마 학부생(a baby undergrad)

대한민국 서울 · [연락처](#)



ASR from Scratch

학생 최재원

Division of Language & AI

2025. 12. 10. 12 PM.

한국외대 인문과학관 206호